

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		$(\text{HC}=\text{O})_2^+ / (\text{HCO})_2^+ / \text{H}_2\text{C}_2\text{O}_2^+$ must have + charge do not accept 2+ charge			1	1		
		(v)		moles of ethane-1,2-diol = $\frac{2.00}{62.06} = 0.0322$ (1) this gives 0.0323 mol of ethanedioic acid M_r of ethanedioic acid = $\frac{3.94}{0.0322} = 122$ (1) relative mass of water = $122 - 90 = 32$ (1) $\Rightarrow x = 1.78 / 1.8 / 2$ (1) Alternative method moles of ethane-1,2-diol = $\frac{2.00}{62.06} = 0.0322$ (1) this gives 0.0323 mol of ethanedioic acid $0.0322 \times 90 = 2.90$ g (1) $3.94 - 2.90 = 1.04$ g of water moles of water = $\frac{1.04}{18} = 0.0577$ (1) mole ratio $0.0322 : 0.0577$ $1:1.79 \quad \Rightarrow x = 1.79 / 1.8 / 2$ (1)		4		4	3	